

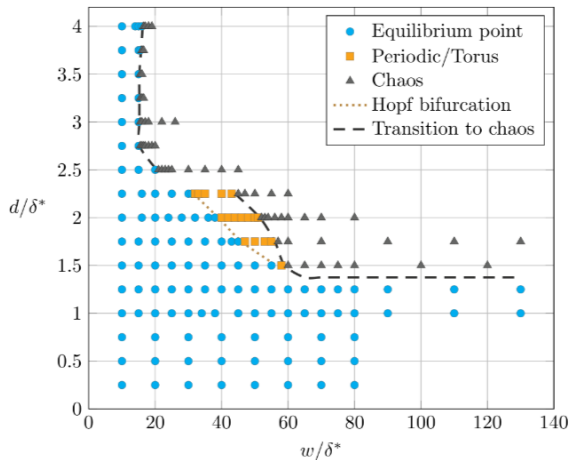
IMPERIAL

$d = 3$ **boundary study**

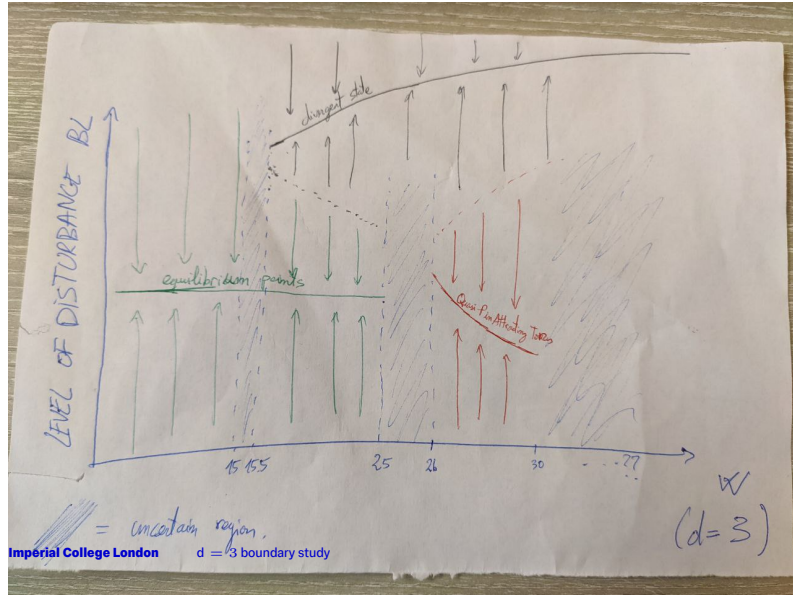
Víctor Ballester
October 3, 2025

Reminder

- We study $d = 3$ bifurcation diagram



Sketch of the bifurcation diagram with the info so far

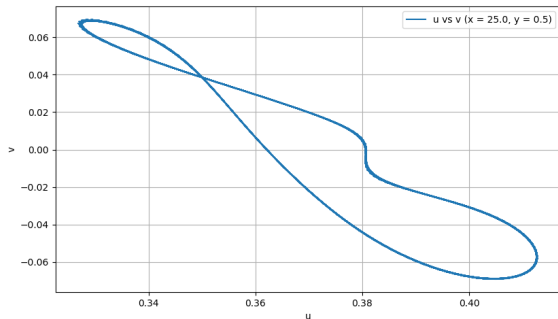


- Bif $w = 15 - 15.5$:
loss of globally attracting state?
- Bif $w = 25 - 26$:
looks Hopf.

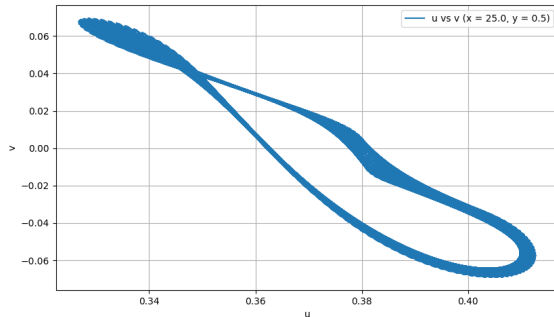
$w = 27$ attractor

- Probably a torus attractor.
- Two different ICs, same convergence. Good sign.

- $dt = 0.001$



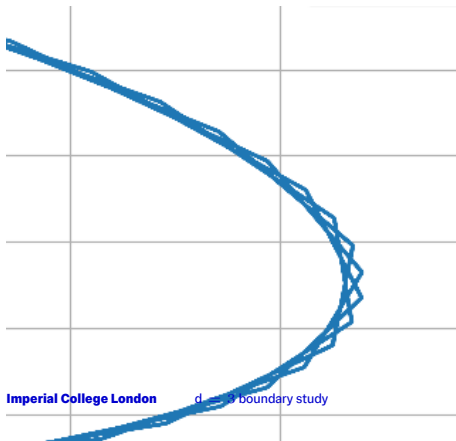
- $dt = 0.004$



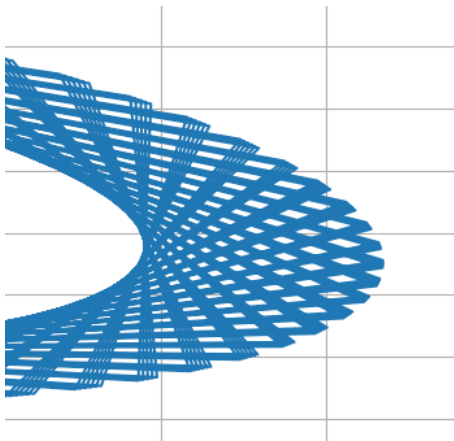
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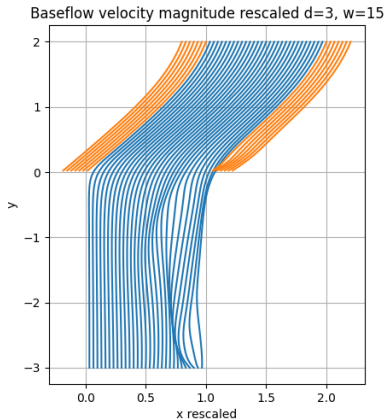
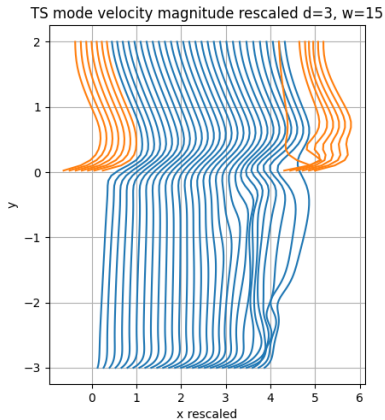


how was the presumably chaotic regime? ($d = 3$, $w = 18$)



sections $d = 3, w = 15$

- TS mode variable is $\sqrt{\langle u^2 + v^2 \rangle}$
- baseflow variable is u



sections $d = 1.25, w = 50$

- TS mode variable is $\sqrt{\langle u^2 + v^2 \rangle}$
- baseflow variable is u

